Cardiff School of Computer Science and Informatics

Coursework Assessment Pro-forma

Module Code: CM1205
Module Title: Architecture and Operating Systems
Lecturer: Dr Yipeng Qin
Assessment Title: Individual Portfolio
Assessment Number: 1
Date Set: 29th January 2024 (Monday, Spring Week 1)
Submission Date and Time: by 9th May 2024 (Thursday, Spring Week 12) at 9:30am
Feedback return date: 4 ***working*** weeks / 20 ***working*** days after submission date

If you have been granted an extension for Extenuating Circumstances, then the submission deadline and return date will be later than that stated above. You will be advised of your revised submission deadline when/if your extension is approved.

If you defer an Autumn or Spring semester assessment, you may fail a module and have to resit the failed or deferred components.

If you have been granted a deferral for Extenuating Circumstances, then you will be assessed in the next scheduled assessment period in which assessment for this module is carried out.

If you have deferred an Autumn or Spring assessment and are eligible to undertake summer resits, you will complete the deferred assessment in the summer resit period.

If you are required to repeat the year or have deferred an assessment in the resit period, you will complete the assessment in the next academic year.

As a general rule, students can only resit 60 failed credits in the summer assessment period (see section 3.4 of the academic regulations). Those with more than 60 failed credits (and no more than 100 credits for undergraduate programmes and 105 credits for postgraduate programmes) will be required to repeat the year. There are some exceptions to this rule and they are applied on a case-by-case basis at the exam board.

If you are an MSc student, please note that deferring assessments may impact the start date of your dissertation. This is because you must pass all taught modules before you can begin your dissertation. If you are an overseas student, any delay may have consequences for your visa, especially if it is your intention to apply for a post study work visa after the completion of your programme.

NOTE: The summer resit period is short and support from staff will be minimal. Therefore, if the number of assessments is high, this can be an intense period of work.

This assignment is worth 100% of the total marks available for this module. If coursework is submitted late (and where there are no extenuating circumstances):

1 If the assessment is submitted no later than 24 hours after the deadline, the mark for the assessment will be capped at the minimum pass mark;

2 If the assessment is submitted more than 24 hours after the deadline, a mark of 0 will be given for the assessment.

Extensions to the coursework submission date can ***only*** be requested using the Extenuating Circumstances procedure. Only students with ***approved*** extenuating circumstances may use the extenuating circumstances submission deadline. Any coursework submitted after the initial submission deadline without *approved* extenuating circumstances will be treated as late.

More information on the extenuating circumstances procedure and academic regulations can be found on the Student Intranet:

https://intranet.cardiff.ac.uk/students/study/exams-and-assessment/extenuating-circumstances
https://intranet.cardiff.ac.uk/students/study/your-rights-and-responsibilities/academic-regulations

By submitting this assignment you are accepting the terms of the following declaration:

I hereby declare that my submission (or my contribution to it in the case of group submissions) is all my own work, that it has not previously been submitted for assessment and that I have not knowingly allowed it to be copied by another student. I declare that I have not made unauthorised use of AI chatbots or tools to complete this work, except where permitted. I understand that deceiving or attempting to deceive examiners by passing off the work of another writer, as one's own is plagiarism. I also understand that plagiarising another's work or knowingly allowing another student to plagiarise from my work is against the University regulations and that doing so will result in loss of marks and possible disciplinary proceedings[1].

[1] https://intranet.cardiff.ac.uk/students/study/exams-and-assessment/academic-integrity/cheating-and-academic-misconduct

Assignment

Your assignment takes the form of a ***structured portfolio*** detailing how you completed each task delivered throughout the module. The purpose of this assignment is for you to reflect on what you've learned during the teaching activities. You will be able to reflect on your initial understanding of topics including, but not limited to, the main components of a computer system and their functionality, assembly language, and operating systems, and engage in a self-reflection of how the exposure to the class material changed your initial understanding. The portfolio is split into **10** sections corresponding to the 10 weeks and the content is semi-structured: you must complete a section for each week like a diary/blog, yet the content itself is expected to include your reflection on both *the learning materials* **AND** *your discussion with peers during the seminar (if applicable)*. Thus, each section is split into **2** parts:

- *Part 1: Reflection on Learning Materials.* This part must include a discussion of topics covered in asynchronous learning activities and perhaps a concise description of tasks completed and how each contributed to your learning. You must cite all the external materials you used, e.g. links to youtube videos, references to academic articles.
- ***Part 2: Reflection on Seminar Discussions (if applicable).*** This part must include a reflection on what you discussed during the seminar. For example, you can start by showing your understanding of the topic before the seminar, and then summarize other students' ideas on the topic, and finally reflect on what you learned from the discussion and what conclusions you have achieved.

When writing your portfolio, you should write each week as a self-contained entry, but you can reference entries from other weeks to ensure all the material from the first section of your learning portfolio are connected. This will allow you to write on a specific topic covered each week, drawing from both academic and professional industries material, spanning text and audio-visual formats, while showcasing your broad grasp of the module learning material.

There is no right or wrong way to approach the learning portfolio. If you follow the instructions, you have creative freedom in terms of which content you choose to include in your work. For example, you may choose to include sketches, brief literature surveys, code examples in Part 1 and mind-mapping or brainstorming figures in Part 2.

Template document

A template document is provided. Please see learning central for details and a link to download the template. ***You must use this template as the basis for your portfolio submission.*** In addition to the template, to help keep you on the right path when writing, here are some suggestions to keep in mind:

- Writing style: Long sentences with complex prose should be avoided.

- Bullet points are acceptable but should not be overused. Apply them appropriately in moderation.
- Stick to the topic: it is easy to lose track and start writing about nonrelated or tangentially related concepts when discussing your work. Keep on point and maintain the focus of your writing on what you have done and what you have learned.
- Active voice: Write in an active tone, in the first person. Avoid sentences like "A literature review was completed covering…" and instead use "I completed a literature review covering…".

Learning Outcomes Assessed

All learning outcomes specified in the module description are assessed in this assignment.

Criteria for assessment

Credit will be awarded against the following criteria.

As stated previously, this assignment is graded as either a **PASS** or a **FAIL**. Portfolios which meet the following criteria may receive a **PASS** mark:

- Contain an entry for each week of scheduled module activities. Note that *you do not have to complete "Part 2: Reflection on Seminar Discussions" if it is not applicable,* for example, a seminar is not scheduled that week. ***Note that you MUST email the lecturer and apply for an exemption if you cannot attend the seminars (e.g., disabilities).***
- Demonstrate clearly your participation in module activities, and how they shaped your learning and understanding of the module.
- Reference your own work. Activities completed in seminars should be discussed from your perspective, *detailing how/what you contributed to and learned from* the discussion.
- Contain enough detail when describing tasks e.g., what the task was, how you managed to complete it, what the outcomes were, and citations to external materials.

You may receive a **FAIL** mark if:

- There is at least one missing entry of the weekly scheduled module activities.
- Insufficient demonstration of your participation in module activities, and how they shaped your learning and understanding of the module.
- Little reference on your own work but descriptions of what your group did.
- Lacking details when describing tasks.

To make it clearer, examples of high-quality/low-quality entries are included in the appendices for your reference.

Feedback and suggestion for future learning

Feedback on your coursework will address the above criteria. Feedback and provisional marks will be returned 4 ***working*** weeks / 20 ***working*** days after submission date via

LearningCentral. Further feedback is available via email/MS Teams meetings upon request and the availability of the lecturer.

Feedback from this assignment will be useful for all assessments involving reflective writing and the final year project.

Submission Instructions

The table below shows the files and formats expected for submission of your portfolio.

Description		Type	Name
Portfolio	**Compulsory**	One PDF (.pdf) or Word file (.doc or .docx)	[student number]-portfolio.(doc or docx) or [student number]-portfolio.(pdf)

This assignment is graded as either a **PASS** or a **FAIL**. Please note there is a strict **11-page limit (Cover Page included), and font should be Calibri size 11**. Appendices and references do **not** count towards this limit.

Any deviation from the submission instructions above (including the number and types of files submitted) *may result in a mark of **FAIL** for the assessment*.

Staff reserve the right to invite students to a meeting to discuss coursework submissions

Support for assessment

Formative feedback on the submission is available, upon request, from the lecturer during Spring Week 1 – Week 6. This feedback will indicate the quality of your submission and indicate a PASS or a FAIL grade.

Support for the programming elements of the assessment will be available in the lab classes of this module.

Appendices

Examples of high-quality entry:

Reflection on Learning Materials (Week 1)

This was my first week with CM1205 Architecture and Operating Systems. This week was mainly an introductory week, so we did not do a seminar discussion. However, during our first seminar, we did do a little test in which we were placed in break out rooms to understand how it will be like in future weeks. The lecture materials for this week were an insight into the history of computer development. I have always found the development of computers fascinating because we now have everything we could need on a tiny little laptop where before we needed huge machines the size of a whole basement to do minor calculations.

My favourite form of early computing to learn about must be the powerful abacus (Bellos, 2012). I was so amazed by the ability that the young girl, shown in the referenced video, had in using the tool to add such large numbers with units in the trillions, especially as I do not understand how to use it myself. From the lecture material for this week, the examples we were given on the different methods of calculations included the powerful abacus, multiplications using Napier's bones (1617), and the Slide rule (1621). All the different methods that were invented in order to do calculations are incredible and it made me realise how much we really take modern calculators and computers for granted, especially since calculators are very common nowadays.



Figure 1 - This image showcases the powerful abacus.

After learning about the progress of calculators (which were referred to as computers in the past), I learnt about the people and their contributions to the development of computers. This went from the famous Charles Babbage, also known as the father of the computer, to Augusta Ada King, one of the first computer programmers and the first to recognise the full potential of computers, to Alan Turing, and finally to George Robert Stibitz, the first to propose the idea of something 'digital'. Finally, we covered hardware implementation. Having done Von Neumann architecture in our first module CM1101 Computational Thinking, I was already familiar with it, but it was good to have a refresher. In order to further familiarise myself with the Von Neumann architecture, specifically the fetch-execute cycle, I watched a YouTube video, "The Fetch-Execute Cycle: What's Your Computer Actually Doing?" (Scott, 2019). This video helped better my understanding of how the components within the control unit (CPU), such as the accumulator and instruction register, work together with the RAM to execute an instruction through the fetch-execute cycle.

Reflection on Seminar Discussions (Week 8)

This week we were tasked with creating an infographic of the registers in an 8086 microprocessor. Within this task, we were to act as graphics designers in a MOOC company and the infographics to be created were for the CM1205 module. An infographic is "a visual representation of information or data" (Nediger, 2020). When it comes to understanding the 8086 registers, I think that a visual representation is the best way around it – though, I may be biased because I tend to be a visual learner and learn better when presented with diagrams and images. Infographics can be used for several reasons, but in this case, they are required in order to explain a complex process.

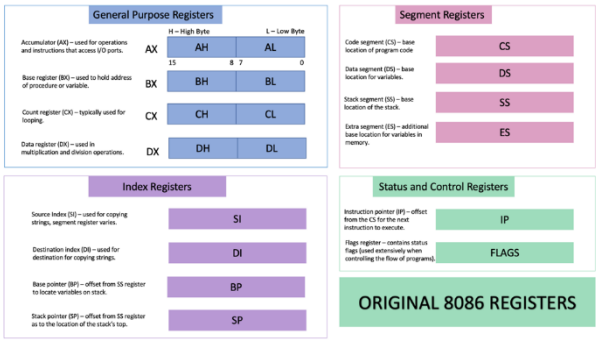


Figure 2 - the infographic I created for the week 5 seminar.

Upon joining the seminar this week, each of my peers, including myself shared our created infographics and then we made comments on one another's. I found that we had all created similar infographics, whether it be Microsoft publisher, Word or PowerPoint. Before beginning, we discussed where we had created our designs and I mentioned that upon researching the topic, I found that there are many tailored websites for this specific area of design. However, we all opted for the simpler options. Upon starting, Simon showcased his

infographic, and I was quick to point out that perhaps colour coding it may make it easier on a viewer's eyes and allows each type of register to be classified by colour. My peers all agreed to this comment. I found it to be a common theme that we all tried to keep the writings minimal in order to make it more approachable and perhaps further explanations would be placed within the MOOC itself. Jasmine pointed out that it was noteworthy that most of us tried to follow the layout of the 8086 register and I agreed with this because it, again, makes it easier on the viewers eyes in terms of visualisation. The main goal was to be brief in explanation and create a visual experience. When one classmate had written quite a lot of text, my classmates gave him some constructive criticism recommending that he decrease the text content which I agreed with! Through reading the other descriptions my classmates had written, it inspired me to refine mine and to attempt to decrease some text a little. It also interested me to investigate the different types of registers again, specifically returning to a previous source and rewatching the video in order to refresh my memory.

Examples of low-quality entry:

Reflection on Learning Materials (Week 2)

This week I completed the week two slides. The slides went through basic computer architecture and the lower levels of the computer level hierarchy (logic gates, binary…). I have not previously learned about decoders and their role in the ALU. I later watched a couple of videos discussing neural networks, as it was briefly mentioned in the lecture. From the first video I understood how forward and backward propagation calibrated the "weights" to give more accurate results. The second video was a part of a tutorial series on using python to make neural networks. It shed light on the mathematics that occurred in the "hidden layers" and what methods to use when implementing it in python.

Reflection on seminar (Week 3)

This week's seminar activity revolved around a scenario where we are required to compare two different CPU based on their specifications. The first CPU being the intel core I5-8265U and the other i7-3517u.

From their names I could immediately see that the I7 processor is from a much older generation. Meaning the i5 would be the better option. Someone mentioned that the I5 had 4 cores compared to 2 for the i7. The obviously makes a big difference in performance. I mentioned that this can be quantified by the fact that the i5 has a clock speed of 3.9 Ghz compared to 3 GHz for the i7.

It was concluded that these arguments would be sufficient to convince the customer. I think comparing CPUs is quite straightforward given the ample online information.